Generating covariate-dependent left truncation times under reverse-time proportional hazards

This note explains how the left truncation time Q is generated in the simulations of `causalDR` (`sim_data_trunc()`) and of `truncAC` (the function `gen()` in `src/gen2.R`, used for the ATE simulation of Wang et al. 2025, hereafter Wang et al.). Both are special cases of one construction, indexed by a baseline distribution function G_0 supported on an interval $[Q_{\min}, Q_{\max}]$. We explain why the construction yields a proportional hazards model on the reversed time scale, what $Q_{\min}$ and $Q_{\max}$ control, and how they interact with positivity and with the estimation of the truncation distribution.

Throughout, $X = (A, Z^\top)^\top$ denotes the treatment and the covariates, T is the event time subject to left truncation (for semi-competing risks $T = \min(T_1, T_2)$), and a subject is observed only if $Q < T$. Let $G(q \mid x) = P(Q \le q \mid X = x)$ denote the conditional distribution function of Q in the population before truncation, and let $\eta^\top x$ be a linear predictor, possibly including an intercept η_0.

1 Reverse-time hazards and the reverse proportional hazards model

The reverse time hazard of Q given $X = x$ is

$$\bar\lambda(q \mid x) = \lim_{h \to 0^+} \frac{P(q - h < Q \le q \mid Q \le q, X = x)}{h} = \frac{g(q \mid x)}{G(q \mid x)},$$

where g is the density of Q, and the reverse cumulative hazard is $\bar H(q \mid x) = \int_q^\infty \bar\lambda(s \mid x)\, ds$. For continuous G, since $\bar\lambda(s \mid x)\, ds = d \log G(s \mid x)$ and $G(\infty \mid x) = 1$, we have $\bar H(q \mid x) = -\log G(q \mid x)$, i.e., $G(q \mid x) = \exp\{-\bar H(q \mid x)\}$, which mirrors the relation $S = \exp(-H)$ in forward time. The reverse proportional hazards model assumes $\bar\lambda(q \mid x) = \bar\lambda_0(q) \exp(\eta^\top x)$, which is equivalent to $\bar H(q \mid x) = \bar H_0(q) \exp(\eta^\top x)$, and hence to

$$G(q \mid x) = G_0(q)^{\exp(\eta^\top x)}, \qquad G_0(q) = \exp\{-\bar H_0(q)\}, \tag{1}$$

where G_0 is a baseline distribution function. Model (1) is the definition of reverse proportional hazards that we use below; it remains meaningful when G_0 has an atom.

The reverse proportional hazards model is a Cox model in forward time after reversing the time axis. For any constant τ at or above the upper end of the support of Q, let $W = \tau - Q$. Then $P(W > w \mid x) = P(Q < \tau - w \mid x)$, and the hazard of W at w equals the reverse time hazard of Q at $\tau - w$. Therefore W given X follows a Cox model with baseline hazard $\bar\lambda_0(\tau - w)$ and regression coefficient η, and this holds for every choice of τ; the constant τ only fixes the origin of the reversed time scale. Moreover, in the observed data Q is right truncated by T, so W is left truncated by $\tau - T$: a subject is at risk at w if $\tau - T < w \le \tau - Q$, i.e., $Q \le \tau - w < T$, which is the risk set

of the backwards martingale $\bar{M}_Q$. This is why both packages estimate G by fitting a Cox model to the data $(\tau - T, \tau - Q]$, with T replaced by X_1 among the subjects with T observed, and inverse probability of censoring weights.

2 A general inverse distribution function construction

Let G_0 be any distribution function supported on $[Q_{\min}, Q_{\max}]$, and let $G_0^{-1}(v) = \inf\{q : G_0(q) \geq v\}$ be its generalized inverse, so that $G_0^{-1}(v) \leq q$ if and only if $v \leq G_0(q)$.

Proposition 1. *Let $U \sim \text{Uniform}(0,1)$ be independent of X and of the potential event times, let $V = U^{\exp(-\eta^\top X)}$, and let $Q = G_0^{-1}(V)$. Then Q given X follows the reverse proportional hazards model (1), and Q is independent of the potential event times given X.*

Proof. For $q \in [Q_{\min}, Q_{\max}]$,

$$P(Q \leq q \mid X = x) = P\{V \leq G_0(q) \mid X = x\} = P\{U \leq G_0(q)^{\exp(\eta^\top x)}\} = G_0(q)^{\exp(\eta^\top x)},$$

where the second equality raises both sides of $U^{\exp(-\eta^\top x)} \leq G_0(q)$ to the power $\exp(\eta^\top x) > 0$. Conditional independence holds because Q is a function of X and of U, and U is generated independently of everything else. □

In words, $V = G(Q \mid X)$ is the conditional probability integral transform of Q: we first draw $G(Q \mid X)$, which is uniform, and then invert $G(\cdot \mid X)$. This is the line `G0_Q = runif(N)^exp(-lp)` in `sim_data_trunc()`. The conditional independence of Q and the event times given X is the conditional independent truncation assumption.

Two remarks on η. First, in the Cox model for $W = \tau - Q$ an intercept η_0 is not identifiable, since $G_0^{\exp(\eta_0 + \eta_1^\top x)} = \{G_0^{\exp(\eta_0)}\}^{\exp(\eta_1^\top x)}$ and $G_0^{\exp(\eta_0)}$ is another baseline distribution function; in the simulation, however, η_0 changes the distribution of Q and is a convenient way to tune the truncation rate. Second, a larger $\eta^\top x$ makes the exponent $\exp(-\eta^\top x)$ of U smaller, so V is closer to 1 and Q is stochastically larger; subjects with a larger linear predictor are therefore more likely to be truncated.

3 The uniform baseline used by Wang et al.

Wang et al. take the baseline to be the uniform distribution on $[Q_{\min}, Q_{\max}]$,

$$G_0(q) = \frac{q - Q_{\min}}{Q_{\max} - Q_{\min}}, \qquad Q_{\min} \leq q \leq Q_{\max},$$

so that by Proposition 1

$$Q = Q_{\min} + (Q_{\max} - Q_{\min})\, U^{\exp(-\eta^\top X)}, \qquad G(q \mid x) = \left\{ \frac{q - Q_{\min}}{Q_{\max} - Q_{\min}} \right\}^{\exp(\eta^\top x)}. \tag{2}$$

Their code generates Q through the reversed time $Q_2 = \tau - Q$ instead:

$$Q_2 = Q_{2,\min} + (Q_{2,\max} - Q_{2,\min})\{1 - U^{\exp(-\eta^\top X)}\}, \qquad Q = \tau - Q_2,$$

with $Q_{2,\max} = \tau - Q_{\min}$ and $Q_{2,\min} = \tau - Q_{\max}$. Substituting these expressions gives $Q = Q_{\max} - (Q_{\max} - Q_{\min})\{1 - U^{\exp(-\eta^\top X)}\}$, which is (2). Hence τ cancels and plays no role in the distribution

of Q. The description "$\tau - Q$ follows a proportional hazards model with the baseline hazard of a uniform distribution" follows from Section 1: when $\eta^\top x = 0$, $W = \tau - Q$ is uniform on $[\tau - Q_{\max}, \tau - Q_{\min}]$, whose hazard is $\lambda_0(w) = 1/(\tau - Q_{\min} - w)$, and the hazard of W given $X = x$ is $\lambda_0(w)\exp(\eta^\top x)$.

The parameters $Q_{\min}$ and $Q_{\max}$ are the lower and upper ends of the support of Q: $G(q \mid x) = 0$ for $q < Q_{\min}$ and $G(q \mid x) = 1$ for $q \geq Q_{\max}$, for every x. They allow the truncation window to be moved, e.g., $Q_{\min} > 0$ when all subjects enter the study late, or narrowed, e.g., $Q_{\max}$ small relative to the range of T for light truncation. Since $E(U^c) = 1/(1+c)$ for $c > 0$,

$$E(Q \mid X = x) = Q_{\min} + \frac{Q_{\max} - Q_{\min}}{1 + \exp(-\eta^\top x)}.$$

In the ATE simulation of Wang et al., $Q_{\min} = 0$, $Q_{\max} = \tau = 5$, and $\eta^\top x = -0.6 + 0.6a + 0.4z_1 + 0.2z_2$, so that, e.g., $E(Q \mid A = 0, Z = 0) = 5/\{1 + \exp(0.6)\} = 1.77$ and $E(Q \mid A = 1, Z = 0) = 2.5$. With the uniform baseline Q is continuous, and there is no probability mass at $Q_{\min}$.

4 A baseline with a point mass at the lower end

The function `sim_data_trunc()` in `causalDR` uses the baseline

$$G_0(q) = p_0 + (1 - p_0)\frac{q - Q_{\min}}{Q_{\max} - Q_{\min}}, \qquad Q_{\min} \leq q \leq Q_{\max},$$

with $Q_{\min} = 0$ and $Q_{\max}$ given by the argument `q_max`. The generalized inverse is $G_0^{-1}(v) = Q_{\min}$ for $v \leq p_0$ and $G_0^{-1}(v) = Q_{\min} + (Q_{\max} - Q_{\min})(v - p_0)/(1 - p_0)$ for $v > p_0$, so that, by Proposition 1, Q follows the reverse proportional hazards model with

$$P(Q = Q_{\min} \mid X = x) = G(Q_{\min} \mid x) = p_0^{\exp(\eta^\top x)}.$$

The uniform baseline of Wang et al. is the special case $p_0 = 0$. The point mass represents subjects who are followed from the time origin, i.e., incident rather than prevalent subjects; such subjects are never truncated, since $Q_{\min} < T$.

Other familiar constructions can be checked in the same way. If $W \sim \text{Exp}\{\lambda\exp(\eta^\top x)\}$ given $X = x$ and $Q = \max(\tau - W, 0)$, then for $0 \leq q \leq \tau$

$$G(q \mid x) = P(W \geq \tau - q \mid x) = \exp\{-\lambda e^{\eta^\top x}(\tau - q)\} = G_0(q)^{\exp(\eta^\top x)}, \qquad G_0(q) = \exp\{-\lambda(\tau - q)\},$$

which is again of the form (1), with a point mass $\exp\{-\lambda\tau e^{\eta^\top x}\}$ at 0. In contrast, generating W from the same exponential distribution conditional on $W \leq \tau$, or generating Q itself from an exponential distribution with rate $\exp(\eta^\top x)$, does not give a reverse proportional hazards model, because $-\log G(q \mid x)$ is then not of the form $\bar{H}_0(q)\exp(\eta^\top x)$; these constructions can be used to misspecify the model for G.

A baseline G_0 without an atom at $Q_{\min}$ requires $G_0(Q_{\min}) = 0$, i.e., the reverse cumulative hazard $\bar{H}_0$ must diverge at $Q_{\min}$, so that $W = \tau - Q$ has a bounded support ending at $\tau - Q_{\min}$ with a hazard that diverges there. The uniform baseline has this property, whereas the exponential construction above, whose support is unbounded before capping, cannot avoid the atom.

5 Forward-time proportional hazards for the event times

The same idea generates event times from a proportional hazards model in forward time,

$$S(t \mid x) = P(T > t \mid X = x) = S_0(t)^{\exp(\eta^\top x)}, \qquad H(t \mid x) = H_0(t)\exp(\eta^\top x), \tag{3}$$

where $S_0 = \exp(-H_0)$ is a baseline survival function, with the survival function in place of the distribution function. Let $S_0^{-1}(v) = \inf\{t : S_0(t) \le v\}$ be the generalized inverse of S_0, so that, since S_0 is non-increasing and right-continuous, $S_0^{-1}(v) \le t$ if and only if $S_0(t) \le v$.

Proposition 2. *Let $U \sim \text{Uniform}(0,1)$ be independent of X, and let $T = S_0^{-1}\{U^{\exp(-\eta^\top X)}\}$. Then T given X follows the proportional hazards model* (3)*.*

Proof. For $t \ge 0$,

$$P(T \le t \mid X = x) = P\{U^{\exp(-\eta^\top x)} \ge S_0(t)\} = P\{U \ge S_0(t)^{\exp(\eta^\top x)}\} = 1 - S_0(t)^{\exp(\eta^\top x)},$$

so that $S(t \mid x) = S_0(t)^{\exp(\eta^\top x)}$. □

In words, we first draw $S(T \mid X)$, which is uniform, and then invert $S(\cdot \mid X)$. Since $E = -\log U$ follows the standard exponential distribution, the construction can equivalently be written as $T = H_0^{-1}\{E\exp(-\eta^\top X)\}$, i.e., the cumulative hazard $H(T \mid X)$ is standard exponential, which is the method of Bender et al. (2005) for simulating Cox models. For example, the exponential baseline $H_0(t) = \lambda t$ gives $T = -\log(U)\exp(-\eta^\top X)/\lambda$, the Weibull baseline $H_0(t) = \lambda t^\nu$ gives $T = \{-\log(U)\exp(-\eta^\top X)/\lambda\}^{1/\nu}$, and the uniform baseline $S_0(t) = 1 - t/\tau$ on $[0,\tau]$ gives $T = \tau\{1 - U^{\exp(-\eta^\top X)}\}$. In contrast to the reverse-time construction, a larger $\eta^\top x$ now makes $U^{\exp(-\eta^\top x)}$ closer to 1 and hence T stochastically smaller, since a larger hazard leads to earlier events. A location shift $T_{\min} + S_0^{-1}\{U^{\exp(-\eta^\top X)}\}$ keeps the proportional hazards structure, with all hazards equal to zero on $[0, T_{\min})$.

The two constructions are mirror images of each other. By Section 1, Q follows the reverse proportional hazards model (1) if and only if $W = \tau - Q$ follows the forward model (3) with the baseline survival function $S_0(w) = G_0\{(\tau - w)-\}$, and for the uniform baseline the forward draw $W = \tau\{1 - U^{\exp(-\eta^\top X)}\}$ is exactly the reversed-time draw $Q_2 = \tau - Q$ in the code of Wang et al. with $Q_{\min} = 0$ and $Q_{\max} = \tau$. In the semi-competing risks generator of `sim_data_trunc()`, the latent event time $T_{\min} + \Lambda_0^{a,-1}\{-\log(U_1)/2\}$ is of this form with $H_0 = \Lambda_0^a$ and the constant multiplier $\exp(\eta^\top x) = 2$; the covariates enter only through their correlation with the latent (U_1, U_2), which is why the Cox models for the transition rates given (A, Z) are misspecified in that design.

6 Positivity and the choice of $Q_{\min}$ and $Q_{\max}$

The estimators weight the observed subjects by $1/G(T \mid A, Z)$, so $G(T \mid A, Z)$ must be bounded away from zero; Wang et al. assume that $T \ge \tau_1$ and $Q < \tau_2$ almost surely, $G(\tau_1 \mid A, Z) \ge \epsilon$ and $1 - F(\tau_2 \mid A, Z) \ge \epsilon$ for some $\epsilon > 0$, where F is the distribution function of T given (A, Z).

The lower end $Q_{\min}$ must lie below the lower end of the support of T. Under the uniform baseline and $T \ge T_{\min}$,

$$G(T \mid x) \ge G(T_{\min} \mid x) = \left\{\frac{T_{\min} - Q_{\min}}{Q_{\max} - Q_{\min}}\right\}^{\exp(\eta^\top x)},$$

which is bounded away from zero if and only if $T_{\min} > Q_{\min}$ and $\eta^\top x$ is bounded. If instead T can be arbitrarily close to $Q_{\min}$, e.g., $T_{\min} = Q_{\min} = 0$, then $G(T \mid x) \to 0$ as $T \to Q_{\min}$ and the weights are unbounded. There are two ways to secure positivity. Wang et al. shift the event time, $T = 1 + \text{Weibull}$, so that $T_{\min} = \tau_1 = 1 > Q_{\min} = 0$ and $G(T \mid x) \geq 0.2^{\exp(\eta^\top x)}$. The point mass of Section 4 achieves the same without shifting T, since $G(T \mid x) \geq G(Q_{\min} \mid x) = p_0^{\exp(\eta^\top x)}$. For semi-competing risks, shifting $T = \min(T_1, T_2)$ amounts to shifting both T_1 and T_2 by τ_1, i.e., all transition hazards vanish on $[0, \tau_1)$; Cox models for the transition rates remain correctly specified, since their baseline hazards are simply zero on that interval.

The upper end $Q_{\max}$ plays the role of τ_2. A subject with Q close to $Q_{\max}$ is observed only if $T > Q$, so G is identified on its whole support only if $P(T > Q_{\max} \mid x) > 0$. The upper end also controls the amount of truncation: by the conditional independence of Q and T,

$$P(Q < T \mid X = x) = E\{G(T- \mid x) \mid X = x\},$$

which decreases as $Q_{\max}$ increases with $Q_{\min}$ fixed, because $G_0(q)$ then decreases for every q, and as $\eta^\top x$ increases. In the ATE simulation of Wang et al. about 23% of the generated subjects are truncated.

7 Implications for the estimation of G

With a continuous baseline such as the uniform one there are no ties among the reversed-time event times $\tau - Q$, and the Cox model for $W = \tau - Q$ described in Section 1 is fitted without complications.

With a point mass at $Q_{\min}$, all subjects with $Q = Q_{\min}$ have their reversed-time event at $w = \tau - Q_{\min}$, and the risk set at that time, $\{Q \leq Q_{\min} < T\}$, consists of exactly these subjects. The true reverse time hazard at $Q_{\min}$ is $P(Q = Q_{\min} \mid Q \leq Q_{\min}, X = x) = 1$ for every x, so the exact partial likelihood contribution of this tied event time is 1 and carries no information about η. The Breslow approximation replaces it by

$$\frac{\prod_{i \in \mathcal{D}} \exp(\eta^\top x_i)}{\left\{\sum_{j \in \mathcal{D}} \exp(\eta^\top x_j)\right\}^{|\mathcal{D}|}},$$

where $\mathcal{D}$ is the set of subjects with $Q = Q_{\min}$, and this factor is maximized at $\eta = 0$ (the Efron approximation has the same dependence on η); it acts as a penalty that shrinks the estimate of η towards zero and does not vanish as the sample size grows. The information carried by $Q = Q_{\min}$ is that no reversed-time event occurs before $w = \tau - Q_{\min}$, since $G(Q_{\min} \mid x) = \exp\{-e^{\eta^\top x} \bar{H}_0(Q_{\min})\}$ and $\bar{H}_0(Q_{\min})$ only involves the reverse time hazard above $Q_{\min}$. The fit should therefore treat the subjects with $Q = Q_{\min}$ as censored at the end of the reversed time scale, i.e., use the event indicator $\mathbb{1}(Q > Q_{\min})$, and set $G(Q_{\min}- \mid x) = 0$, so that the atom $dG(Q_{\min} \mid x) = G(Q_{\min} \mid x)$ is retained in the integrals with respect to the backwards martingale $\bar{M}_Q$.

References

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